

SPARK-IL: Spectral Retrieval-Augmented RAG for Knowledge-Driven Deepfake Detection via Incremental Learning

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15:00-16:30 | Forum (floor -2)

A0 PORTRAIT

WHY SPARK-IL?

The open-world challenge

- Generator-specific spatial artifacts do not transfer reliably to unseen synthesis pipelines.
- Frequency-domain statistics are more consistent across GAN, face-swap and diffusion families.
- Static parametric classifiers cannot absorb new generators without costly retraining and forgetting.

GOAL An adaptive detector whose decision is grounded in retrieved spectral evidence - not a fixed global boundary.

CORE CONTRIBUTIONS

- Dual-path spectrum: raw RGB pixels + high-level ViT-L/14 features.
- Four-band KAN-FFT blocks with band-specific mixture-of-experts.
- Multi-head cross-attention aligns pixel-level and semantic spectral anomalies.
- Milvus cosine retrieval + majority voting for evidence-based inference.
- Replay, distillation and EWC preserve prior knowledge while the database expands.

NON-PARAMETRIC CORRECTION AT INFERENCE

EXPERIMENTAL SETUP

UniversalFakeDetect

19 generators

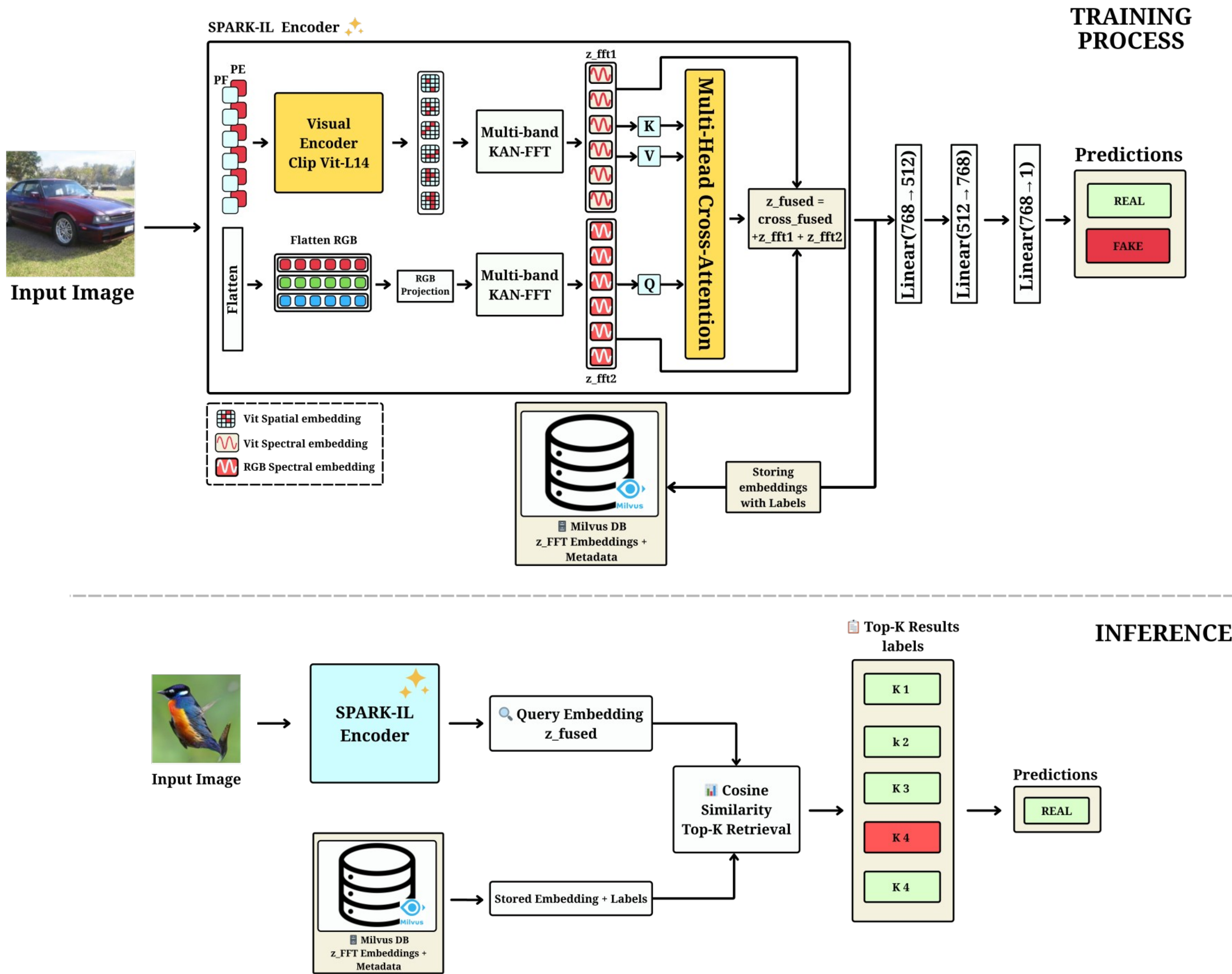
Train: ProGAN only

- Backbone: partially frozen CLIP ViT-L/14; only the last two transformer blocks are adapted.
- Input 224 x 224; Adam, 10 epochs, batch size 64; NVIDIA Quadro RTX 5000 (16 GB).
- Two KAN-FFT blocks per path; 12-head cross-attention; cosine search in a local Milvus database.

Evaluation spans GANs, deepfakes, low-level manipulations, perceptual-loss models and diffusion generators.

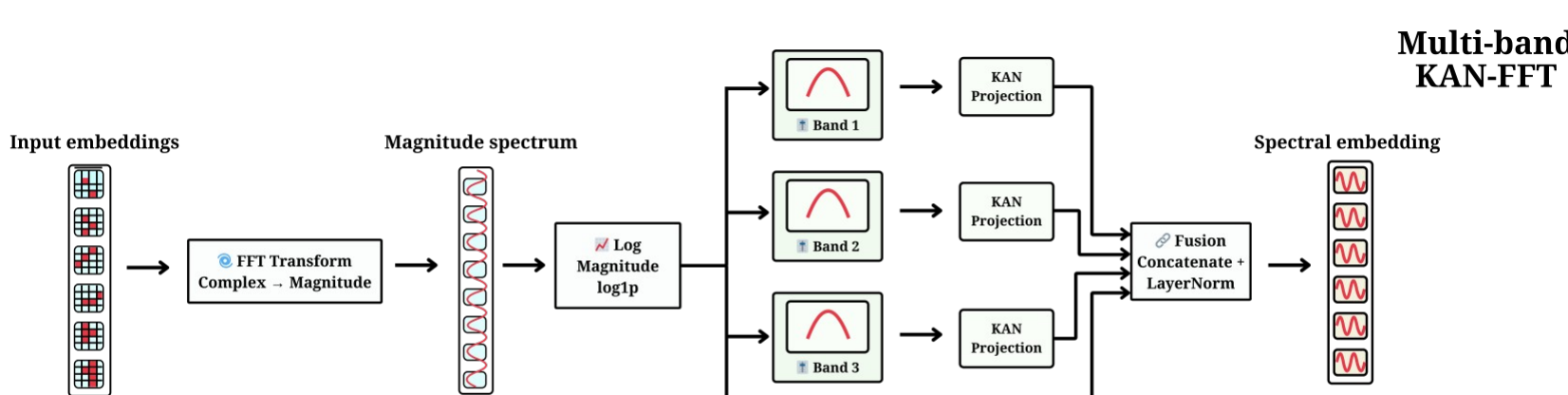
Metric: mean accuracy (mAcc)

SPARK-IL ARCHITECTURE - TRAINING AND RETRIEVAL INFERENCE



HOW IT WORKS

1 MULTI-BAND SPECTRAL PROCESSING



- FFT -> log-magnitude spectrum -> four disjoint frequency bands (d/4 = 192).

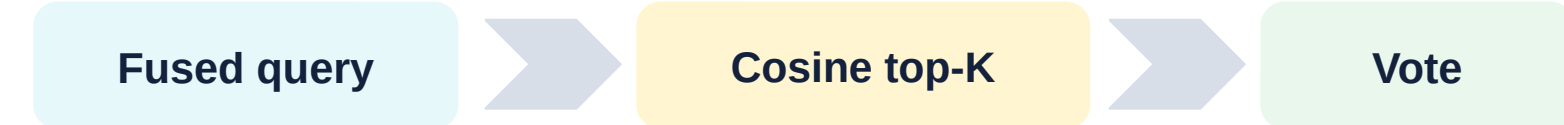
- Each band uses a KAN mixture-of-experts with four expert transformations.

2 CROSS-ATTENTION FUSION

- RGB spectral embedding is the query; ViT spectral embedding provides keys and values.

- Residual fusion preserves both paths: $h_{fused} = h_{cross} + 0.2 z_{RGB} + 0.2 z_{ViT}$.

3 RETRIEVAL + INCREMENTAL LEARNING



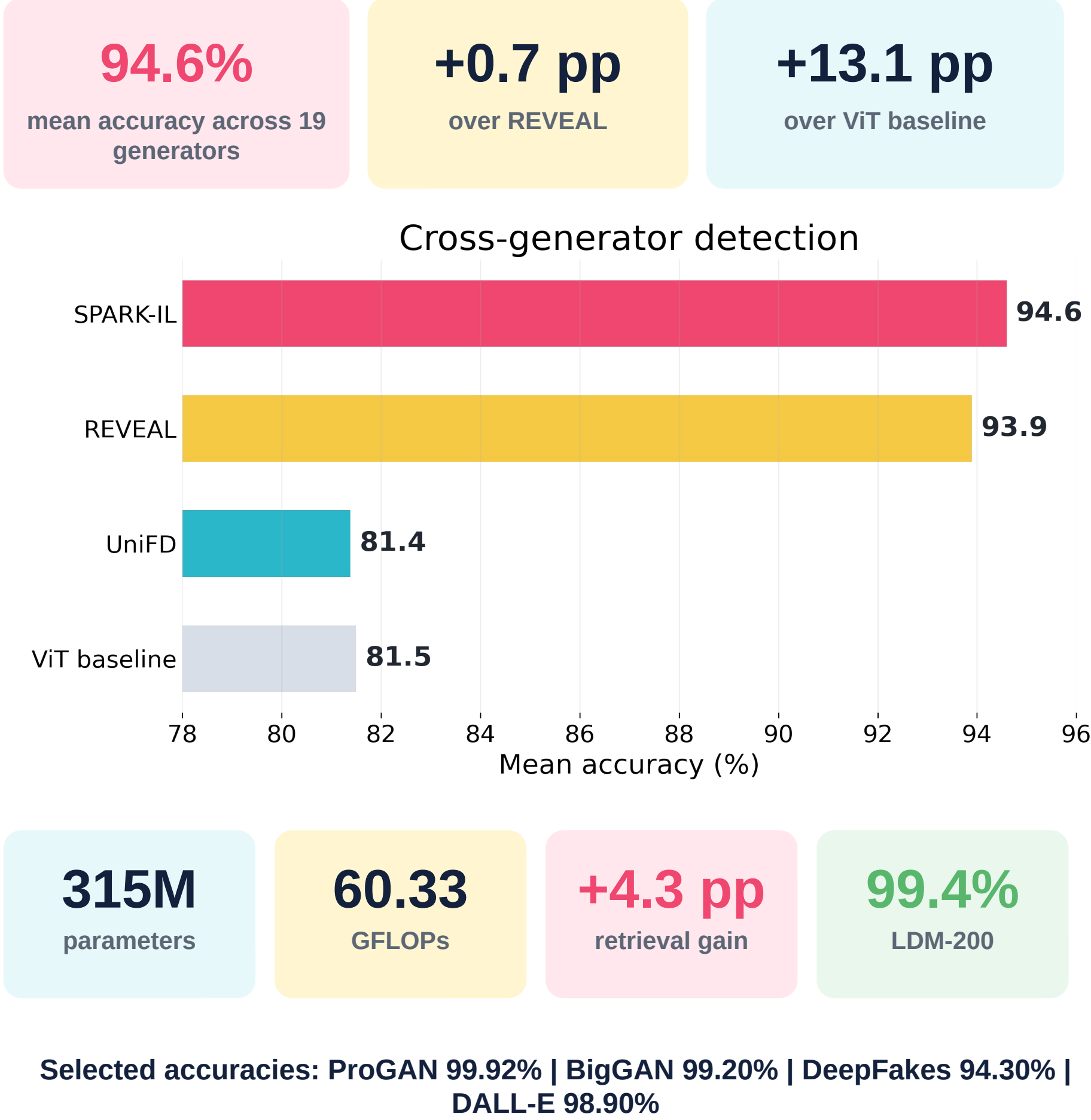
- Predictions use local majority voting over stored labels; the paper evaluates up to K = 20.

- New generators are learned with experience replay, embedding/logit distillation and EWC.

- New fused embeddings are appended to Milvus, expanding knowledge without overwriting prior exemplars.

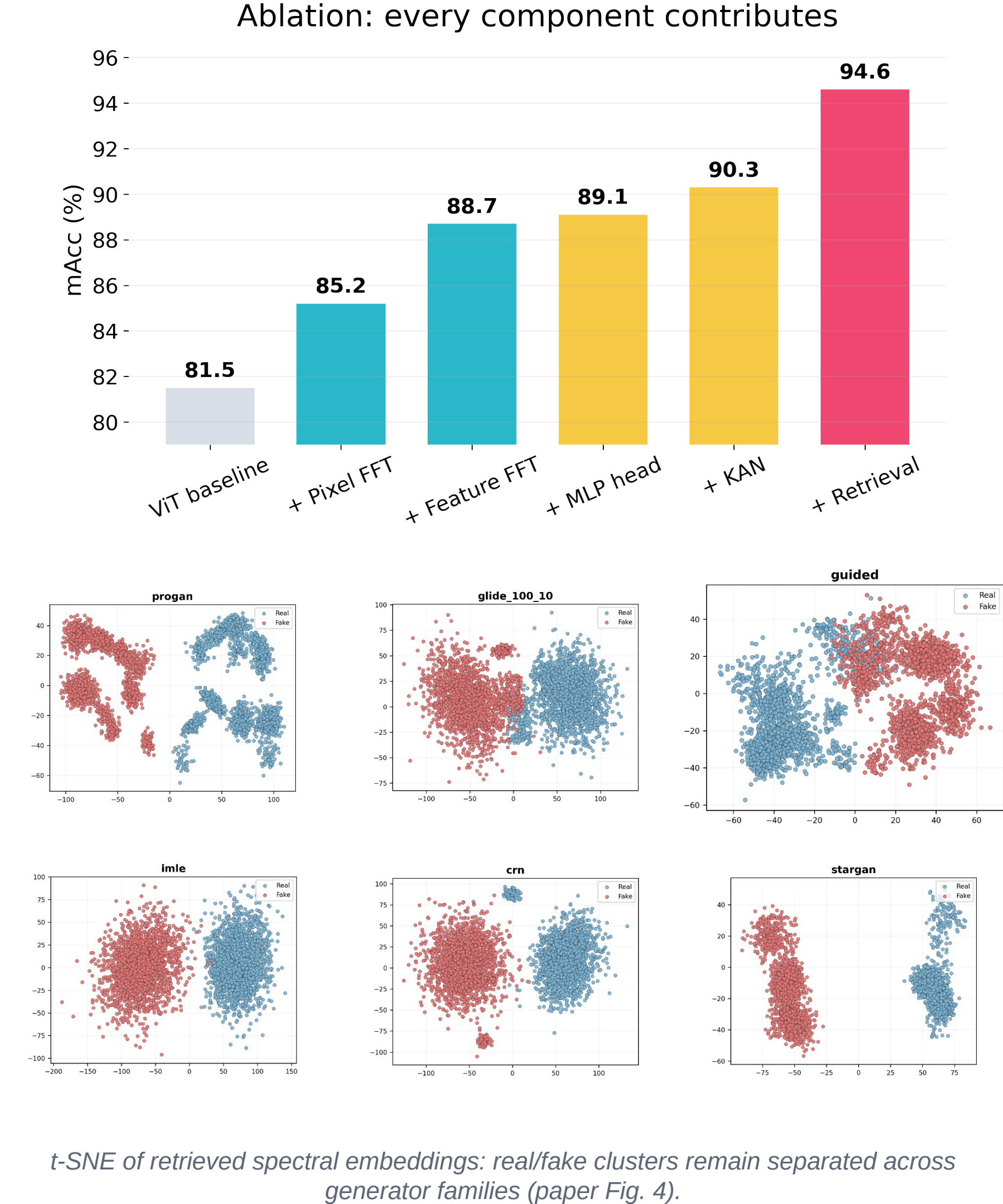
ADAPT MODEL + EXPAND MEMORY

STATE-OF-THE-ART RESULTS

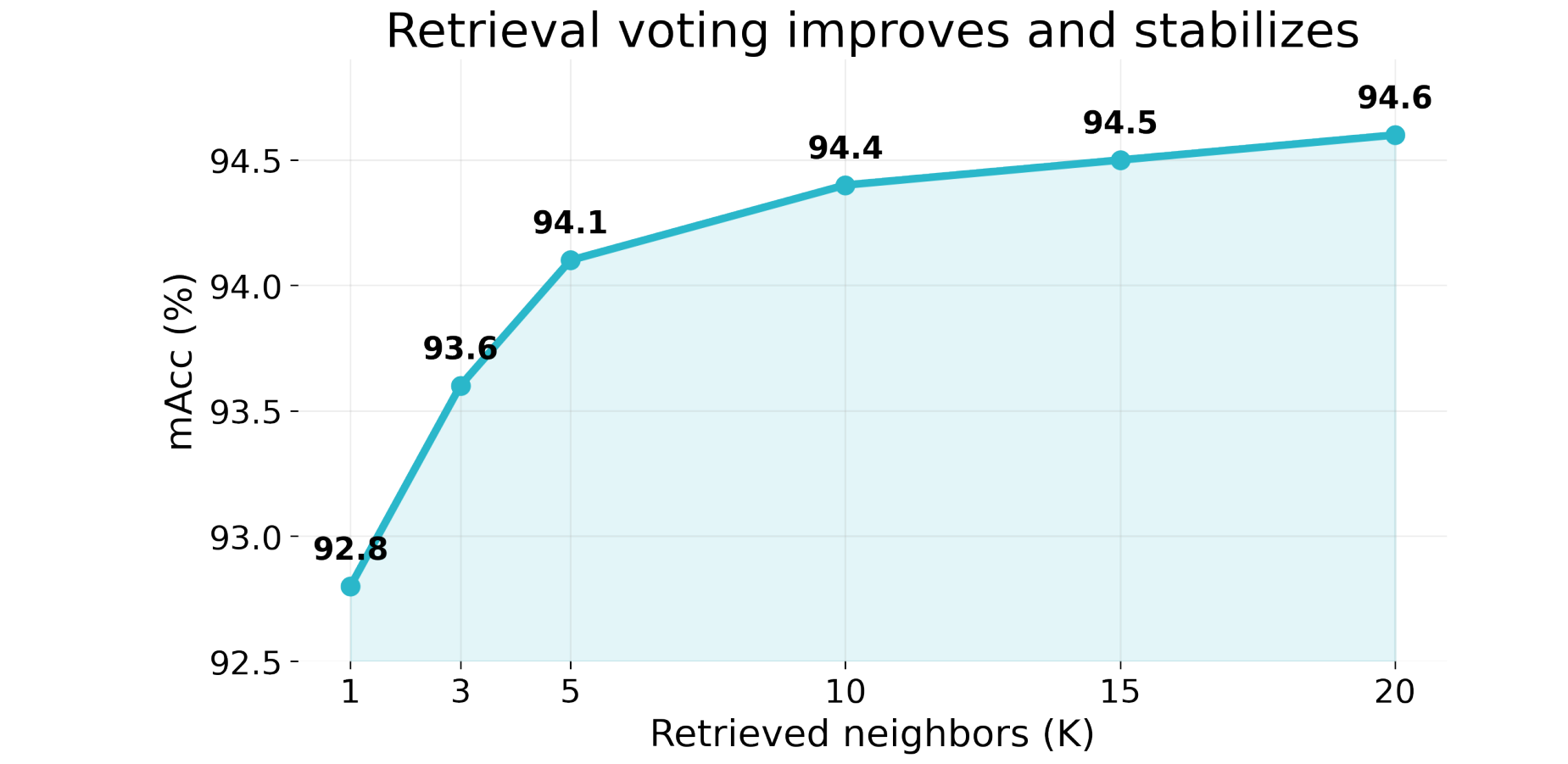


Largest single improvement: retrieval-based voting adds +4.3 pp over KAN without retrieval.

ABLATION + EMBEDDING QUALITY



RETRIEVAL ANALYSIS + TAKEAWAYS



WHAT TO REMEMBER

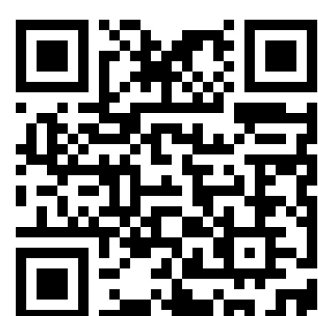
- Frequency cues generalize better than generator-specific spatial artifacts.
- Dual pixel/feature spectra are complementary; cross-attention aligns them.
- Retrieval makes predictions local, evidence-grounded and easy to update.
- Database expansion supports cumulative knowledge as new generators emerge.

LIMITATIONS & NEXT STEPS

Global FFT is weaker for spatially localized artifacts. Future work: patch-level retrieval, adaptive frequency partitioning and multi-scale spectral analysis.

SPARK-IL combines dual-path spectral modeling, retrieval-augmented inference and incremental learning to deliver 94.6% cross-generator accuracy.

Paper: arXiv:2604.03833 | Code: github.com/HessenUPHF/SPARK-IL
References: Ojha et al., CVPR 2023; Keita et al., DFW 2025; Kirkpatrick et al., PNAS 2017.



READ THE
PAPER
arXiv



CODE RELEASE
GitHub